

# Utsav Acharya

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A **Computer Engineering** student with experience in **Machine learning**, **Backend systems**, and **Applied AI research**. Built and deployed data-driven solutions across **computer vision**, **LLM systems**, and **forecasting** through research collaborations and real-world projects.

## WORK EXPERIENCE

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### Student Trainee

04/2025 – 11/2025

NAAMI

Research trainee working on applied computer vision systems in agricultural and biological monitoring domains within a physics laboratory setting.

- Developed **deep learning-based disease detection models for rice crops**, improving classification accuracy to **~93% on curated field datasets** with a **Raspberry Pi-based edge AI application**
- Designed and trained **computer vision pipelines for poultry behavioral monitoring**, enabling automated detection of abnormal movement patterns with **real-time inference capability**.
- Processed and annotated **10,000+ agricultural and livestock image samples** for supervised model training.
- Experimented with **CNN-based architectures (ResNet, EfficientNet variants)** for improved robustness under low-quality field imagery conditions.

### Backend Engineer and Data Scientist

11/2023 – 03/2025

Omdena 🌐

Collaborated with global teams on applied AI projects focused on environmental monitoring and agricultural analytics.

- Built ML pipelines for **Combatting Illegal Deforestation in Romania** 🌐 achieving **~88% segmentation accuracy in prototype models**.
- Contributed to geospatial analysis workflows for **Mapping Seagrass Meadows** 🌐 **using remote sensing data and satellite imagery classification techniques**.
- Improved dataset preprocessing efficiency by **~40%** through optimized ETL pipelines and automation scripts for **Agriculture Analysis on Kenya** 🌐.
- Worked in cross-functional teams of **50+ contributors**, coordinating model development, data preprocessing, and evaluation workflows.

## ACADEMIC PROJECTS

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### MEDIWO — LLM-Powered Medical Workflow Optimization 🌐

AI-assisted healthcare platform designed to optimize outpatient department (OPD) workflows by automating patient intake, reducing consultation overhead, and generating structured clinical summaries for doctors prior to consultation.

- Engineered a full-stack platform using **React, TypeScript, Tailwind CSS, FastAPI, and MongoDB**, supporting separate patient and doctor dashboards for streamlined clinical interaction.
- Developed an adaptive **AI-driven intake chatbot** that collected and structured patient symptoms, reducing manual intake time by **~65%**.
- Built a **RAG-based clinical summarization pipeline** using **Llama, LangGraph, and Chroma**, achieving **92% summary relevance accuracy** during internal evaluation.
- Integrated an **OCR-based medical report digitization module**, enabling automated extraction from **500+ uploaded diagnostic documents**.
- Designed a **human-in-the-loop feedback framework** that logged physician corrections as reward signals, improving summary precision by **18% across iterative retraining cycles**.

### Compliance Guide — Intelligent Compliance Analysis System 🌐

AI-powered compliance intelligence platform built to simplify complex policy documents and automate security framework analysis for faster gap identification and remediation.

- Built a **document intelligence pipeline** using **locally hosted Llama models** for automated policy summarization and contextual compliance interpretation.
- Integrated a **RAG-based retrieval system** that improved compliance recommendation accuracy from **76% to 95%**.
- Developed a **comparative evaluation framework** using dual LLM inference pipelines to benchmark response consistency across compliance scenarios.

- Processed and indexed **1,200+ compliance documents** across security standards including ISO 27001 and SOC 2.
- Reduced average compliance analysis time from **4 hours to under 20 minutes per document set**.

### Volt Vision — Household Energy Demand Forecasting for Nepal

Machine learning-based forecasting system for predicting household energy demand and electricity pricing trends in Nepal.

- Processed and analyzed **30,000+ household energy consumption records** using **Pandas and NumPy** for large-scale demand pattern analysis.
- Trained and optimized **XGBoost forecasting models**, achieving **14% lower MAE** than baseline regression approaches.
- Implemented **SARIMAX and ARIMAX models** for multivariate time-series forecasting of household demand and tariff fluctuations.
- Built end-to-end **MLOps deployment pipelines** using **ZenML and MLflow**, enabling experiment tracking and reproducible model versioning.
- Improved forecasting reliability for seasonal demand spikes with **91% prediction accuracy during peak usage periods**.

## EDUCATION

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### Bachelor's in Computer Engineering

*Pulchowk Engineering Campus*  
Average Percentage: 82.4

2023 – present  
Kathmandu, Nepal

### High School

*Capital College and Research Centre*  
GPA: 3.91 + 3.73

2021 – 2023  
Kathmandu, Nepal

## SKILLS

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**Languages** — Python, C, C++, JavaScript, SQL

**Data Analysis** — NumPy, Pandas, SciPy, Regex, Tableau

**AI Systems** — Scikit Learn, TensorFlow, Pytorch, LangChain, LangGraph, RAG, Prompt Engineering

**Backend & Deployment** — FastAPI, Flask, Docker, AWS, Linux, MLflow

**Mathematics** — Linear Algebra, Probability & Statistics, Numeical Optimisation with Calculus

## CERTIFICATES

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|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| • AWS Solutions Architect Associate         | • Google Advanced Data Analytics                 | • IBM Data Science Professional Certificate      |
| • Stanford Machine Learning Specialization  | • Deep Learning.AI Deep Learning Specialization  | • Tensorflow Developer Professional Certificate  |

## LEADERSHIP & EXTRACURRICULARS

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### Data Research Council For Students

*Core Member*

11/2022 – 12/2023

- Mentored **200+ students** in **intermediate Python**, providing guidance and support to help them improve their skills.
- Managed **GitHub** repositories, **social media** accounts, and **Discord** servers, ensuring that these platforms were up-to-date and effectively engaged with our community.

### Together We Learn

*Co founder*

03/2022 – 08/2023

- Co-founded and led an organization of **500+ members**
- Conducted **5+ webinars** on programming-related topics, sharing knowledge and insights with a broad audience.
- Wrote **10+ newsletters** focused on **data science** and **programming** in general, providing valuable information and updates to **200+ subscribers**.

### Kathmandu Valley Leo Club

*Volunteer*

01/2023 – 07/2025

- Participated in different volunteering activities from KVLC Pulchowk.